CARRT Assistive Arm CSE Project — Weekly Status Report # 5

Current Report Period: Weeks #4-5 ***(End-of-Period Report)***

1. Goals for the Current Period?

Carry-Over Goals (unaccomplished from prior weeks/reporting periods)

1. **Gazebo ADL Simulation, Week #3 –** [✓ | ✗]– Successfully test at least 2 of 3 ADLs using the simulated arm in Gazebo, with observable arm movement and task completion.
    a. ***Problem:*** Arm movement in Gazebo wasn't achieved by the end of week 3. Time was spent on installation of simulator, URDF loading, and environment configuration, with little time to complete and test joint actuation sequences.
    b. ***Approach:*** Implement a ROS2 joint trajectory node to drive the arm, verify basic movement, then build ADL specific poses for each task. Target at least two tasks with repeatable movements by the end of the period.

New Goals

2. **Requirements Document** – [✓ | ✗] – Consolidate project requirements into one PDF regarding discussed goals, assumptions, concepts, and criteria. Each member reviews assigned sections, then complete the total document and submit with Week 4
3. **ROS2 / Kortex Develop Environment** – [✓ | ✗] – Create a ROS2 Python coding environment with their library and confirm functionality on the main simulation machine. Install kortex_api following Kortex documentation. Troubleshoot package / dependency issues and confirm Python node's control over arm commands.
4. **QR Code Recognition** – [✓ | ✗] – Develop a functional base layout for camera object recognition using OpenCV and AprilTags that can detect a small set of QR tags and publish the tag's pose into ROS2. Create a perception node prototype that can use OpenCV camera feed for AprilTag detection, print a set of tags and verify stable detection / pose output. Define finalized object / locations lists and QR placement rules.
5. **3D Printed Modeling** – [✓ | ✗] – Design at least two basic 3D models for objects used during simulated / physical testing. CAD and print essential objects/props needed for early testing with consistent measurements.
6. **UI Development / Voice Command Integration** – [✓ | ✗] – Investigate UI strategies for app-based voice command integration with ROS2 and Python. Provide at least two viable solutions and select one. Compare alternatives (ROS2 native UI vs tablet/voice integration methods). Decide on a direction depending on the project needs and timeline, to move forward with next Period.

2. Goals Accomplished in the Period?

1. **Task #1: Gazebo ADL Simulation –** [Accomplished / ***Partially***]: The simulation was confirmed functional and arm movement through ROS2 Python nodes achieved. Skeleton behaviors for the ADLs were implemented. Full repeatable execution of the tasks was not achieved, because tasks scripts were not finalized by EoP.
    a. ***Completed By:*** **Oliver (4 hrs), Jadon (3 hrs)**
2. **Task #2: Requirements Document** – [***Accomplished*** / Partially]: The requirements document was successfully submitted alongside the Week 4 report. It captures goals, assumptions, main concepts, and success criteria for the project.
    a. ***Completed By:*** **Entire Team (~20 min each)**
3. **Task #3: ROS 2 / Kortex Environment Development –** [***Accomplished*** / Partially]: The kortex issues were identified and resolved, incorrectly written links in install packages caused delays in development. The environment is now fully functional for simulation and coding.
    a. ***Completed By:*** **Oliver (5 hrs), Jadon (3 hrs)**
4. **Task #4: QR Code Recognition –** [***Accomplished*** / Partially]: A functional QR detection layout was developed with OpenCV and AprilTags. Multiple tags were printed and successfully detected with pose data verified returning to ROS2. Object list and placement rules finalized.
    a. ***Completed By:*** **Allision (3 hrs)**
5. **Task #5: 3D Printed Modeling –** [***Accomplished*** / Partially]: basic 3D models were created and used, including a table and a water bottle for ADL testing.
    a. ***Completed By:*** **Oliver (30 min)**
6. **Task #6: UI Strategy –** [***Accomplished*** / Partially]: A ROS2-native UI node was selected as the preferred implementation.
    a. ***Completed By:*** **Brielle (1-2 hrs)**

3. Critical Reflection on Unaccomplished Goals for the Period (to Carry-Over)

1. **Task #1: Gazebo ADL Simulation –** Full ADL execution was not yet achieved. Significant time this period was spent working through major diagnostic issues in installation, which blocked code writing and delated task scripting. Once environment setup was complete, there was little time remaining for refining and testing the ADL sequences. Skeleton behaviors exist but aren't yet fully robust / executable.

a. ***Revised Approach:*** With the environment now stable, the focus will shift to ADL task scripts and we will distribute the scripting across team members. Targeting a 30-50% repeatable success rate for at least 2 tasks before the Design Review presentation.

4. Major Goal List for Next Period (End of Period)

1. **Gazebo ADL Simulation –** Achieve successful, repeatable execution of at least 2 ADL tasks in simulation, with a 30-50% success rate across multiple tests.
    a. ***Approach:*** Use confirmed ROS2 environment for testing. Assign specific ADL tasks to team members. Run test sessions with logged pass/fail results.
2. Design Review Presentation – Prepare and deliver the presentation in class, with a PowerPoint and rehearsed scripts for the Week 6 class presentation.
    a. ***Approach:*** divide slide sections by project component and have at least one team practice prior to presentation.
3. Tablet Wireframes – Design at least two tablet UI wireframes for the ROS2-native UI node. Capture layout, interaction flow, and voice command triggers.
    a. ***Approach:*** Wireframe creation led by Brielle, based on UI strategy selected this period. Share drafts with team before finalization.

5. Work Division for the Team

- *Divisions are specifically noted in Sections 2 and 3. For each team member, by total hour count for the current week:*

Name:	Oliver Gaston	Jadon Orr	Allison Moline	Brielle Ashmeade
Hours Total:	~9.5 hrs	~3.5 hrs	~3 hrs	~1.5 hrs